Effects of prompt style on user responses to an automated banking service using word-spotting

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Two experiments were performed to measure the effects of differing styles of prompt wording in a simulated telephone banking service incorporating a speech recogniser with word-spotting capabilities. It was found that users gave longer input utterances in response to an 'open' style of prompt ('How can I help you?') than in response to a 'closed' prompt which was more specific as to what input was expected, and that when a help facility was offered most users said 'help' straight away. However, no significant difference was found in attitudes to the different versions of the service. Also no attitude difference was found with varying recognition accuracy. This may have been partly due to inadequate vocabulary coverage obscuring the effect of accuracy within the vocabulary, but the results suggest also that small numbers of recognition errors may have little impact on user attitude provided that the intended result of the call is achieved — this is a topic where further research could be valuable.

1. Introduction

With the continual improvement of speech recognition technology, it is now possible to offer automated telephone services where the user is not confined to saying isolated words or digit strings but can give spoken input in a more natural form — including phrases or whole sentences — from which the recogniser can extract the information relevant to the task by means of a continuous speech recognition or word-spotting algorithm.

Given this type of recognition capability, it is of interest to know what style of service prompt wording is appropriate to optimise the overall usability of a service. Prompts giving specific instructions as to what to say may be the best for eliciting easily recognised within-vocabulary input from the user, but may result in a rather unnatural style of dialogue. In contrast, a more open style of prompt wording may seem more natural but result in a higher incidence of unrecognisable or inaccurately recognised inputs. The optimal prompt wording may vary with the accuracy of the recognition technology, with a more open style preferred when the accuracy is high (so that the recogniser can reliably pick out a keyword which may be embedded in a sentence), and a more constraining style preferred when the recogniser is less accurate (in which case it will tend to make substantially more errors on embedded keywords than on isolated ones). As a more general issue, it is useful to know what users will say in response to a given prompt, as this information can be used in the design of recognition vocabularies and grammars.

To explore these issues, an experimental methodology for usability evaluation [1] has been extended to include an acoustically guided 'Wizard of Oz' simulation of a word-spotting recogniser, in which acoustic pattern-matching scores are combined with information from the simulation operator (or 'wizard') as to what was actually said, to arrive at a recognition of the input speech. The results are realistic in that the accuracy is higher for complete isolated keyword utterances than for embedded or incomplete ones.

Using this technique, two experiments have been conducted, with simulated automated telephone banking services, to explore the effects of prompt wording style on users' spoken responses and on their attitudes to the service, and to test the hypothesis that the relative attitudes to different styles of prompt wording will differ according to the accuracy of the word-spotting recogniser.

Section 2 of this paper describes the word-spotting simulation, and sections 3 and 4 report on the two experiments and their findings; some general comments and conclusions are given in section 5.

2. The word-spotting simulation

The use of Wizard of Oz (WOZ) simulations [2], in which a hidden human operator, the 'wizard', takes the place of an automated component such as a speech recogniser, allows the usability of automated dialogue systems with future technology to be investigated experimentally, since the simulation is not restricted to the capabilities of currently available technology.

In the simplest case, the wizard replaces the speech recogniser completely, typing in what the user says (provided that this is within the designated recognition vocabulary), so that the simulated system has a recognition capability, within its vocabulary, equivalent to that of a human listener.

This can be refined by introducing recognition errors automatically between the wizard's input and the system's response. For realism, the distribution of errors across words in the vocabulary should depend on the phonetic similarity of the words, and the overall error rate should vary from one user to another to reflect the 'sheep and goats' phenomenon whereby most recognition errors occur on a minority of the user population.

WOZ simulations used in past research have operated with keyboard input from the wizard, together with a statistical model of the recogniser from which errors can be generated randomly to accord with the general pattern of errors expected from a real recogniser at the designated accuracy [1, 3]. This form of simulation has the limitation that it takes no account of the acoustic quality of a particular input utterance — it uses only the identity of the input (i.e. which word or word sequence, if any, from the vocabulary was spoken) as entered by the wizard. This limitation is particularly serious when simulating word-spotting, since the error rate of a real word-spotting recogniser depends strongly on whether the keyword to be recognised occurs in isolation or embedded in other speech, and on the length and other characteristics of the utterance in the latter case. Therefore, in order to simulate word-spotting technology more realistically, a new form of simulation has been developed which uses the acoustic signal as well as the information entered by the wizard.

2.1 General description

The simulation (Fig 1) combines information from a simple acoustic pattern-matching algorithm with the information entered by the wizard as to the 'right answer', i.e. what the user actually said. The wizard's input is used to enhance the pattern-matching score for the right answer so as to improve the basic performance of the simple algorithm to the desired level of accuracy. The procedure for each input utterance is as follows.

The user's speech is passed into the signal-processing and pattern-matching module, which locates the start and end of the utterance, derives a parametric representation (a sequence of cepstral vectors) and then matches it against a pronunciation network to determine the best-matching recognition hypothesis.

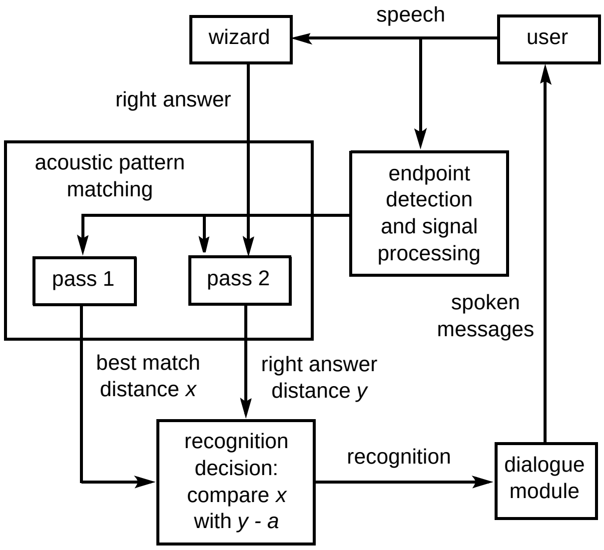

Fig 1 The word-spotting simulation.

At the same time the wizard listens to the input and enters the correct identity of the keyword spoken (ignoring any surrounding speech) — or enters a 'no keyword' code if the utterance does not contain a keyword from the designated active vocabulary.

The pattern matcher then performs a second pass using a pronunciation network which allows only the actually spoken keyword (if any) with arbitrary surrounding speech. The correct-hypothesis score from this second pass is enhanced by a fixed amount (determined in advance to provide the desired level of recognition accuracy) and then compared with the score from the first pass. If the enhanced correct-hypothesis score is better than the first-pass score, the correct recognition is adopted; otherwise, the best-scoring hypothesis identified in the first pass is adopted.

2.2 Details of the acoustic processing

The acoustic pattern-matching component is derived from one originally developed for phonetic segmentation of utterances of known phrases [4] and uses a simple form of hidden Markov model.

For each pattern-matching pass, a pronunciation network is defined which specifies the permitted words or word sequences in terms of a set of acoustic-phonetic units (APUs). Some of the APUs correspond directly to phonemes such as monophthongal vowels, fricatives and nasals, while others correspond to parts of phonemes such as the closure or release portion of a stop consonant. Each APU is represented by a one-state model with a single cepstral centroid vector.

The matching is done using the Viterbi algorithm, augmented with a post-processing form of duration scoring applied at each state transition. The output comprises the best-matching APU sequence permitted by the pronunciation network, with the start and end times of each

APU segment, and an overall distance score. (Only the overall score and the keyword label or labels associated with the APU sequence are used for the purpose of the recogniser simulation.)

For the word-spotting simulation two types of network are used. The vocabulary model allows any word of the current active vocabulary, with optional arbitrary preceding and following speech ('garbage'); it also includes a no-keyword path in which the whole utterance is matched to the garbage model. The word model allows only a single specified keyword, again embedded in optional arbitrary speech. A special case of the word model is the no-keyword model in which the keyword is null and so only a no-keyword path is present.

The model for 'garbage' consists of a loop which contains models for all the APUs in parallel and which can be repeated indefinitely. The APU models used in the garbage modelling have their distance scores systematically increased, by multiplication of the cepstral distance at each frame by a suitable constant, so that a good utterance of a keyword will match the keyword model better than the garbage model. (Without this adjustment, the no-keyword path would always have at least as small a distance as the correct-keyword path.)

In the first pass of the simulation procedure, the vocabulary model is used to find the best-matching hypothesis (of a keyword or no keyword) with the corresponding distance score x. In the second pass, the word model is used to find the distance y for the 'right answer' as entered by the wizard. This is enhanced, i.e. reduced, by a constant amount a (set to give the desired level of recognition accuracy on a calibration database), and the enhanced right-answer score $y-a$ is compared with the first-pass score x to get the simulated recognition decision.

The APU models used in the current implementation have been trained on hand-segmented data from BT's SUBSCRIBER database [5]. Separate models were trained for male and female speakers.

2.3 *Details of the wizard's interface*

This simulation is intended to be used for dialogues in which the active vocabulary varies from one point to another. The keyboard input method is adequate for digit and yes/no recognition, but it is less suitable where the dialogue involves diverse vocabularies, because of the burden imposed on the wizard who must remember at each point what the active words are and which key corresponds to each word. Accordingly a visual interface was introduced, in which the words of the current vocabulary appear on the computer screen and the wizard can select the word spoken (or 'no keyword' if appropriate) using a mouse. The wizard's screen (Fig 2) also shows a graphical display of the progress of the speech data acquisition and

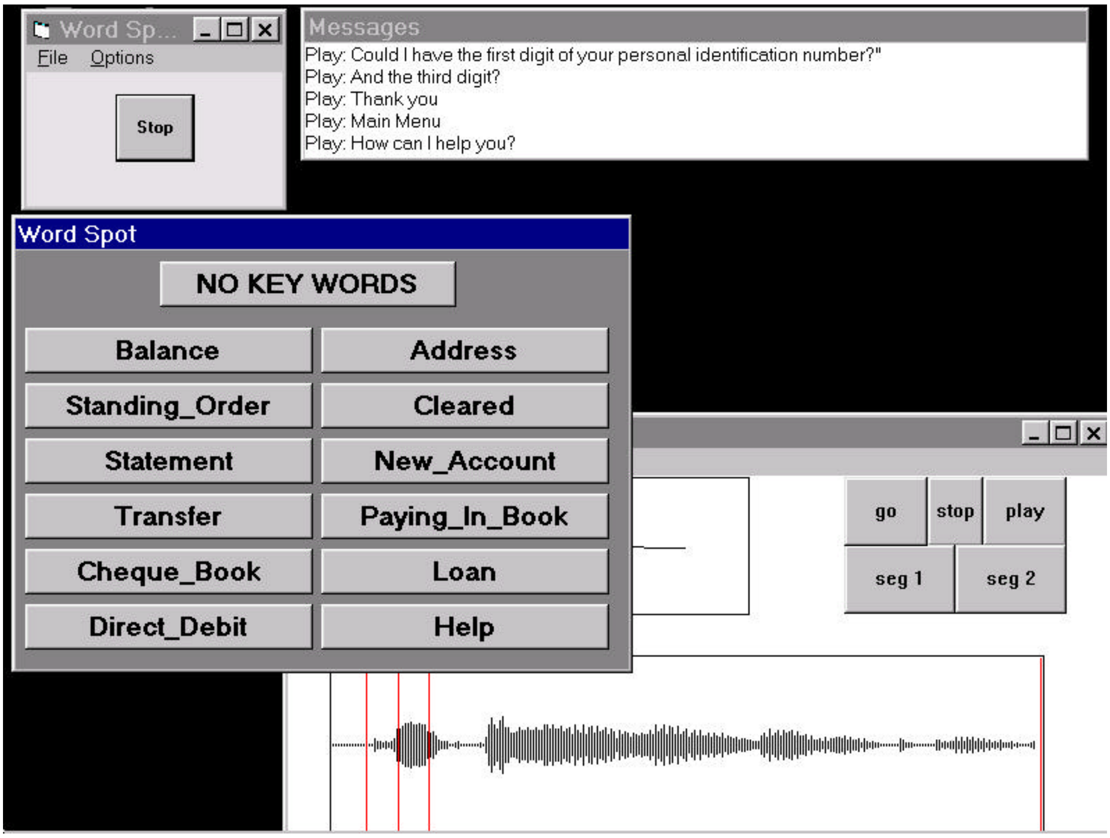

Fig 2 The wizard's interface to the word-spotting simulation.

end-point detection. This enables the wizard to see if the end-point detector has truncated the user's utterance so as to miss the keyword, in which case the wizard should select the 'no keyword' option.

In case of silence, or of input too quiet to trigger the end-point detector, the system times out automatically; the wizard does not need to do anything in this case (and if the wizard does enter anything it is ignored).

If the user's utterance contains more than one keyword when only one is expected, the wizard must decide which to enter. It is recommended that the longest keyword in the utterance should be chosen — this is realistic in that long words tend to be easier to detect automatically in connected speech than short words.

3. First experiment — comparing three prompt styles

In the first experiment carried out with the word-spotting simulation, the main focus was on the wording of a particular prompt, occurring at the point in the dialogue where the user was asked to make a choice from a set of available service options, namely different types of banking transaction. To avoid complications and distractions, the word-spotting recogniser simulation was applied only in this section of the dialogue, and at all other points the wizard's input was used directly (resulting in 100% accurate recognition, apart from any errors by the wizard).

3.1 Banking service dialogue

When a user called the service, the dialogue began with prompts for an eight-digit account number and for two digits selected from a four-digit personal identification number (PIN). The user was asked to speak the numbers — the account number as an eight-digit string, and the PIN digits as isolated words in response to separate prompts. If valid numbers were given, the dialogue proceeded to the 'Get Service' phase. (If invalid numbers were given, a second attempt was allowed.)

In the Get Service phase, the initial prompt was one of the following:

- open: 'Main Menu — How can I help you?'
- mid: 'Main Menu — Which service do you require?'
- closed: 'Main Menu — Please say 'help' or the name of the service you require,'

in each case followed by a tone or 'beep'. The second-level prompt, occurring after a silence or no-keyword response in the open or mid style, was the same as the top-level prompt in the closed style. The second-level prompt in the closed style and the third-level prompt in all cases included a list of the service keywords. The list of keywords was also provided if the user said 'help'. There were 11 service keywords: *balance, standing_order, [mini_]statement, transfer, cheque_book, direct_debit, [change_of_]address, cleared[_cheques], new_account, paying_in_book* and *loan.* (Elements in square brackets were optional; if present, they would tend to improve the chance of correct recognition, since the word model in each case included the long form as well as the short form.) The recognition vocabulary consisted of these service names plus 'help'.

The responses to the service prompts were recognised using the word-spotting simulation. This was calibrated on a small set of data (recorded by members of the research team interacting with a prototype version of the service) to give an agreed level of performance. On the calibration data, the false detection rate for no-keyword utterances was 17%, and for keyword utterances (just over half of which were embedded rather than isolated) the miss rate was 2% and the substitution rate 10%, giving an accuracy of 88%.

The next phase of the dialogue depended on which service keyword was recognised. In the case of a balance or statement request, the appropriate information was simply read out to the user. In the case of a cheque book request, a confirmation message was given ('A cheque book will be sent to you'). In the case of a 'cleared cheques' enquiry, a six-digit cheque number was requested, and the user was told whether that cheque had or had not been cleared. In all other cases, the user was told that that service was currently not available.

After this, the user was prompted to hold the line for another transaction, or else hang up. If the line was still open after a two-second pause, the dialogue looped round to the top-level Get Service prompt.

At every point, up to three attempts at speech input were allowed, with more detailed prompts given at the second and third attempts. In the event of three successive errors (silence or invalid or rejected input), the service terminated the call with an appropriate message ('I'm sorry, there seems to be a problem'). A context-dependent help message was available at every point in the service, and was obtained by saying 'help' in response to the prompt.

3.2 Experiment design and procedure

For this experiment 152 people were recruited from the Edinburgh area through a market research company. Each participant was allocated to one of the three prompt styles. (The design called for 50 participants per prompt style; in fact 52 were obtained for the closed style, and 50 for each of the others.)

Each participant was given a general description of the service and a list of four tasks to be accomplished using it, comprising a balance enquiry, a statement request, a cheque book request, and an enquiry as to whether a specified cheque had been cleared — with the order of the four tasks being varied from one participant to another. The tasks could be completed in a single call, or, if preferred, the participant could hang up between tasks and call the service again. The instructions did not include a list of the service keywords or stipulate whether to respond with single words or with phrases or sentences.

The participants' utterances in response to the service prompts were digitally recorded and subsequently transcribed, since a major aim was to determine what people said in response to each style of prompt. Statistics on the performance of the simulated recogniser were also collected.

After completing their tasks, participants completed a Likert-style attitude questionnaire [6, 7] to provide a subjective evaluation of the usability of the service.

3.3 Results

Analysis of participants' inputs

The participants' responses on first encountering the Get Service prompt are shown in Table 1, analysed by form (to distinguish single words and short phrases from sentences) and keyword content. The numbers in parentheses are the numbers of inputs of the given types containing a valid keyword (service name or 'help'). The 'other' class includes unintelligible or incoherent inputs, and those whose classification as 'word/phrase' or 'sentence' is uncertain because the end-point detector cut off part of the utterance. A classification by content only is shown in Fig 3.

Table 1 Analysis of responses at first encounter with Get Service prompt in the first experiment.

Response type	Closed style		Mid style		Open style	
silence	5		8		3	
help	36	(36)	0	(0)	0	(0)
word/phrase	4	(4)	9	(7)	2	(2)
sentence	1	(1)	12	(8)	25	(13)
other	6	(3)	21	(9)	20	(7)
TOTAL	52	(44)	50	(24)	50	(22)

A major difference is evident between the responses from participants given the closed prompt style and from those given the mid or open style. The closed-style top-level prompt explicitly mentioned the possibility of saying 'help', and 36 of the 52 participants did so at the first opportunity. Having heard the list of keywords, the majority of them then responded with an isolated keyword utterance when reprompted. In the mid and open conditions, in contrast, no one said 'help' initially (since the possibility of doing so had not been mentioned), and most of the first-time inputs either contained a service keyword embedded in other speech (43 out of 100) or contained no keyword (43 out of 100, plus 11 instances of silence). For no-keyword inputs in response to the open prompt, the detected speech was nearly always either a sentence or sentence fragment or else just a filler such as 'er' or 'ah'; both these categories also occurred in the mid condition, but there were also some attempts at guessing keywords such as 'enquiries please'.

Those mid-style and open-style participants who failed to have a keyword recognised at the top level then heard a prompt identical to the top-level prompt in the closed condition. Their responses from that point on were similar to those of the closed-style participants.

On the second, third and fourth tasks, the contrast in response types between the closed condition and the others was less sharp. More of the participants now knew what the keywords were and therefore were able to give valid inputs. However, a strong tendency persisted towards isolated keywords in the closed condition and a balance of isolated and embedded ones in the other conditions.

There were significantly more silences overall in the mid condition (0.9 per participant) than in the open condition (0.3). (A *t*-test on the per-participant silence counts returned $p = 0.003$.) This suggests that an open-ended prompt such as 'How can I help you?' encourages people to say something even if they do not know exactly what the options are, whereas a specific prompt such as 'Which service do you require?' may cause users to remain silent if they are not sure of the exact option name.

The total number of utterances per participant for the four tasks was similar across all three groups — the reduction due to having fewer unrecognised inputs in the closed condition was roughly balanced by the increased number of utterances of 'help'.

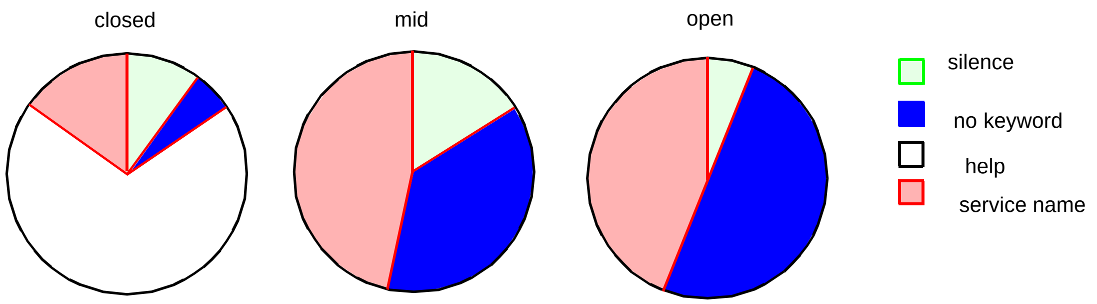

Fig 3 Content of initial responses to the Get Service prompt in the first experiment.

Simulated recogniser performance

The evaluation of the recogniser simulation was complicated by end-point detection errors, which resulted either in the truncation of the keyword spoken or in its complete exclusion from the data passed into the acoustic pattern-matching component. In some cases the wizard heard and entered a keyword but it was excluded by the end-point detector, so that the second pass of the simulation was matching a keyword model against an interval of speech which did not in fact contain the keyword. However, by suitable classification of the inputs and exclusion of such anomalous cases, measures of recogniser performance can be obtained.

For no-keyword utterances correctly entered as such by the wizard, the rate of false keyword detection was 20%, the other 80% being correctly recognised as 'no keyword'.

For all keyword utterances — defined as those for which the wizard entered a keyword, whether or not it was captured by the end-point detector — the recognition accuracy was 89%, with 7% misses (no keyword recognised) and 4% wrong-keyword recognitions. The accuracy was 95% on complete isolated keyword utterances, 89% on complete embedded keywords, and 78% on incomplete keywords. (An incomplete keyword occurred in the case of truncation by the end-point detector, or if the participant said something which was not exactly the designated word, such as 'clearance' for 'cleared', but it was treated by the wizard as an instance of the keyword.) On keywords excluded completely by the end-point detector, the accuracy was 58% — unrealistically high because the simulation had the benefit of the 'right answer' entered by the wizard, who heard the whole utterance rather than just what the end-point detector retained. Apart from this last, anomalous case, the pattern of results seems plausible. The errors on short utterances tended to be no-keyword recognitions, and those on longer utterances to be wrong-keyword recognitions, which seems realistic. Also the performance under experimental conditions was similar to that on the calibration data.

Questionnaire responses

Despite the major differences in the spoken responses to the three versions of the service, the overall attitudes to all three versions were similar as measured by the questionnaire — the mean scores (on a seven-point scale, with 1 being the least favourable evaluation possible, and 7 the most favourable) were 5.22 (closed), 5.19 (mid) and 5.00 (open), and the differences between these were not significant ($p > 0.05$) according to a one-way ANOVA. Five of the individual items in the questionnaire showed significant differences between the conditions — attitude to service voice ($p = 0.04$), informativeness of the service messages ($p = 0.02$), knowing how to select options ($p = 0.001$), usefulness of the help facility ($p = 0.026$) and attitude to time taken by the service to deal with requests ($p = 0.049$). For the first of these, the attitude was higher in the mid condition than in the open and closed conditions; for the others, the closed version of the service scored higher than the open version, while the relative position of the mid version varied.

4. Second experiment — the interaction of prompt style and recogniser accuracy

The main aim of the second experiment was to explore the effects of prompt style and recognition accuracy on the usability of a service with word-spotting, and, in particular, the possible interaction between them. Accordingly, two styles of prompt wording — open and closed — were tested at three levels of recognition accuracy. The hypothesis was that attitudes to the version of the service with the open style would be more favourable, relative to attitudes to the version with the closed style, at a high level of recogniser performance than at a low level, since at the low accuracy level the increase in recognition errors due to embedded rather than isolated keyword input (encouraged by the open prompts) would be greater, counteracting any advantage of the open style in respect of perceived naturalness.

Other aims were to collect further data on what people said in response to different styles of prompting and to extend the testing and calibration of the word-spotting recogniser simulation. This experiment went beyond the first one in several respects:

- three levels of recogniser performance were simulated instead of just one, so as to address the issue of recognition accuracy and its interaction with prompt style,
- each participant made multiple uses of the service, with attitude questionnaires after the first and last uses, to allow attitudes on first use and after repeated use to be distinguished,
- the banking service was extended to include setting up a standing order, a more complex type of transaction requiring the input of several pieces of information,
- the acoustically guided word-spotting simulation was used for all inputs from the user throughout the dialogue, rather than just for the inputs in the Get Service block,
- the priming for the tasks was designed to simulate real banking scenarios more closely by the use of dummy documents — a cheque book (containing several stubs and a complete cheque, as shown in Fig 4), a letter from the beneficiary of the standing order, and a separate sheet of paper bearing the PIN.

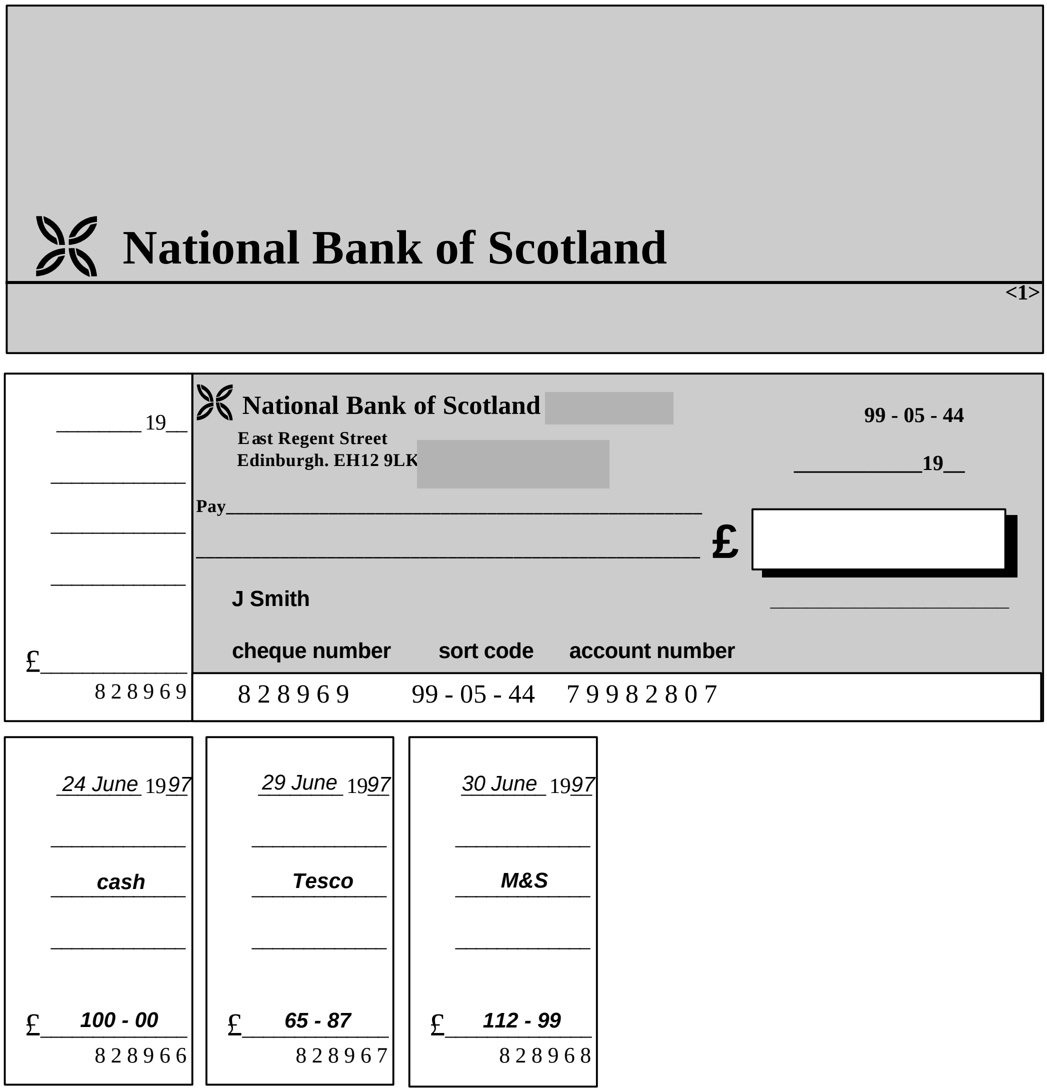

Fig 4 Pages of a simulated cheque book as used in priming for the second experiment.

4.1 *Experiment design and procedure*

For this experiment 120 participants were recruited, none of whom had taken part in the first experiment. They were divided into six groups of 20, with 10 male and 10 female participants in each group, to correspond to the six experimental conditions (i.e. two prompt styles combined with three accuracy levels). The same set of 20 task sequences was used within each group, and, for analysis purposes, the participants were matched across groups for gender and task sequence.

Each sequence consisted of three tasks, to be carried out in separate calls to the service. Each task consisted either of one complex transaction (setting up a standing order) or of two different simple transactions (balance, statement, ‘cleared cheques’ enquiry or cheque book request). The order of the transaction types was balanced across participants, except that the account balance transaction always preceded the statement transaction (for realism, since the balance was included in the statement and so there should be no need to ask for it afterwards).

Usually each task was completed in a single call, but participants were allowed to call the service again if they failed to complete the task on the first attempt. After the first task, and again after the third task, a Likert attitude questionnaire was administered, with the same set of statements as in the first experiment.

All detected input utterances were digitally recorded and transcribed, and statistics were obtained on the performance of the recogniser simulation, as in the first experiment.

4.2 Results

Analysis of participants' inputs

The point where the greatest contrast was expected between the closed and open prompt styles was at the user's response to the first Get Service prompt. In the closed dialogue, this prompt was 'Please say help or say the name of the service you require.' In the open dialogue, it was 'How can I help you?' In each case, it was preceded by the announcement 'Main Menu'.

The pattern of inputs at this point in the dialogue confirms the tendencies found in the previous experiment. Given the closed prompt, many participants said 'help', and a substantial proportion of the other inputs were single words or short phrases such as 'account balance', whereas the great majority of responses to the open prompt consisted of full sentences. The full analysis of responses at the first encounter with the Get Service prompt (in the first use of the service) is shown in Table 2, in which the classification is the same as in Table 1.

Table 2 Analysis of responses at first encounter with Get Service prompt in the second experiment.

Response type	Closed style		Open style	
silence	6		0	
'help'	25	(25)	0	(0)
word/phrase	12	(10)	1	(1)
sentence	7	(4)	52	(45)
other	10	(6)	7	(3)
TOTAL	60	(45)	60	(49)

The 'sentence' class includes a few instances of long phrases which were probably not intended to be simply service names, such as 'standing orders to set up'. In some cases classed as 'other', the participant was apparently asking the experimenter how to respond, or else thinking aloud, since the detected utterance contains phrases such as 'service you require' and 'how can she help me'. While there were slightly more no-keyword utterances in the open case (11, against 9 in the closed case), there were also 6 silences in response to the closed prompt, and so the overall incidence of failure to give a keyword was slightly higher in the closed than in the open condition.

All of the 25 participants who initially requested help managed to give a service name when reprompted, though 5 of them asked for help a second time before doing so. In most cases (21 out of 25), the service-name input contained just the service name, or the service name followed by 'please' or preceded by 'help' (the latter perhaps due to misinterpreting the instruction 'Please say help or say the name of the service you require').

The pattern of inputs at the second level in the open condition (in case of silence or rejection at the first attempt) was similar to that at the top level in the closed condition, and the pattern at the second level in the closed condition was similar to that after a help message — as might be expected from the similarity of the prompts and information messages.

Similar differences between the conditions persisted to a reduced degree at subsequent visits to the Get Service block.

The closed and open conditions differed at two other points in the dialogue, where the user was asked for the name of the payee and the payment frequency for the standing order. The difference in the payee prompts was rather similar to that in the Get Service prompts: 'Who do you want the payments made to?' (open) versus 'Please say 'help' or the name of the organisation you want your payments made to' (closed); and a similar contrast in the patterns of responses was observed, though to a reduced degree since in both conditions most responses were payee names in isolation. The payment frequency prompt in the open condition was 'How often would you like the payments to be made?' and in the closed condition 'Would you like the payments to be made monthly, quarterly, or annually?' The latter elicited fewer embedded-keyword and no-keyword responses and more silences, though the no-keyword and silence differences were small and not statistically significant.

In the other parts of the dialogue the prompts were the same in both conditions. The following features of the responses were noted.

- About 83% of PIN digit responses were isolated digits. Other responses included silence (4%), phrases like 'number 8' (5%), and sentences like 'The second digit is 6' (2.5%).
- When prompted for an eight-digit account number or a six-digit cheque number (in each case, to be read from the cheque book), most participants gave a digit string of the correct length, sometimes preceded by a phrase containing 'account number' or 'cheque number', but there were also some instances where too few digits were entered (either because the participant said the PIN instead of the account number or because the end-point detector truncated the utterance at a pause), and some utterances containing the out-of-vocabulary word 'double'. The rate of correct-length string entry for account numbers was 89%, but for cheque numbers only 80%, with higher rates of both silences and no-keyword responses, evidently reflecting the fact that participants were not told explicitly in the printed instructions that they would have to identify the cheque by number.

- After confirmation prompts (ending ‘Is this correct?’ or ‘Is that correct?’), negative responses were nearly always (51 times out of 55) just ‘no’, but positive responses were more varied, including ‘yes’ and ‘yes that’s correct’ (both of which could be recognised) and also ‘correct’ and ‘that’s correct’ (which were not recognised since the defined vocabulary contained only ‘yes’ and ‘no’).

- Some participants volunteered extra details during the standing order sub-dialogue, for instance giving the payment frequency as well as the amount when prompted for the amount.

- The prompt ‘Which day of the month?’ caused some problems, with at least 10% of participants interpreting ‘day’ as ‘day of the week’, despite the context and the fact that the standing order instructions in the letter gave only the date and not the weekday, and giving replies such as ‘Thursday’ (which of course could not be recognised).

Simulated recogniser performance

As with the first experiment, the transcribed input utterances were classified according to the complete or partial presence or absence of a keyword, and in case of a complete keyword according to whether it was isolated or embedded, and recognition statistics were obtained for each class of input. This was done for each recognition vocabulary in turn, at each calibrated recogniser performance level.

As intended, the error rates for all vocabularies and input classes were zero at the high performance level. (There was one instance where the input speech was too short to match the right-answer model, leading to an ‘infinite’ distance in the second pass of the simulation; but arguably the rejection that occurred in this case was not an error since the keyword was not in the detected utterance and so the wizard should have treated it as a no-keyword input.) Also the error rates were substantially greater at the low performance level than at the intermediate level. A comparison between the results in the experiment and those on the calibration data revealed no systematic difference.

The error rates for all the vocabularies at the intermediate and low performance levels are shown in Table 3. (These results exclude utterances for which the wizard entered a keyword but it was wholly excluded by the end-point detector. This occurred for about 8% of all keyword utterances.) Error rates are shown as percentages of the number of non-excluded keyword utterances, or of the number of no-keyword utterances in the case of false detection rate. Where there were no no-keyword utterances, a dash is shown in the false detection column.

Questionnaire responses

Each participant completed a Likert questionnaire after the first task, and another after the third (last) task. The mean attitude scores for the six groups are shown in Table 4.

Related sample *t*-tests were applied to the within-participant differences from first to third use, for each experimental group separately, and, for each use number, to the pairwise differences between corresponding participants’ attitudes with different accuracies (but the same prompt style) and with different prompt styles (at the same accuracy). Also two-way ANOVAs were performed for first-use attitudes and for third-use attitudes with prompt style and recogniser accuracy level as the independent variables. None of the effects attained significance.

Table 3 Recognition error rates in the second experiment.

Vocabulary	Intermediate accuracy			Low accuracy		
	substitution	rejection	false detection	substitution	rejection	false detection
service names	1%	0%	3%	9%	2%	26%
digits	0%	0%	0%	3%	0%	0%
6-digit strings	9%	2%	0%	16%	2%	0%
8-digit strings	16%	2%	26%	28%	3%	14%
yes/no	0%	0%	0%	0%	0%	0%
payees	2%	2%	0%	1%	1%	0%
amounts	9%	0%	0%	25%	1%	—
payment frequencies	0%	0%	0%	1%	4%	—
months	0%	2%	—	6%	11%	0%
days of month	2%	0%	0%	14%	0%	40%

Table 4 Mean attitude scores in the second experiment.

Condition	First use	Third use
closed, high accuracy	4.93	4.93
closed, intermediate accuracy	5.07	4.99
closed, low accuracy	5.18	5.10
open, high accuracy	5.22	5.14
open, intermediate accuracy	5.09	5.16
open, low accuracy	4.88	4.85

Thus the experiment did not reveal any overall effect of prompt style on attitude, nor did it support the hypothesis of an interaction between accuracy and prompt style in determining attitude. More surprisingly, it did not show any effect of recognition accuracy level on users' attitudes to the service. The apparent lack of an effect of recognition accuracy on attitude is not only intuitively surprising but also in contrast to the results of a previous experiment using a similar attitude questionnaire [6] in which highly significant differences were found between attitude scores for a simple numeric data entry dialogue at 100% accuracy and at lower accuracy levels.

Assuming, as seems likely, that recognition accuracy does have some effect on user attitude, and assuming also, as the previous results suggest, that this can, in principle, be detected by the questionnaire used here, there are several factors which may have contributed to the failure to detect such an effect in this experiment.

- Sample size

  The sample size for each accuracy level (40 participants) was rather less than the 51 per accuracy level (after losses from the initial 64 matched quadruples) in the earlier experiment.

- Complexity

  In this experiment the tasks and vocabularies were more complex and less well-defined from the user's point of view than in the earlier experiment with its simple data entry scenario (giving a credit card number as a sequence of 16 isolated digits). This could be expected to introduce more variability, for instance in participants' choice of words to specify their transactions, resulting in variations in their experience of the service and consequently in their attitude towards it. Also the complications arising in the tasks may have had a distracting effect, making participants less aware of the performance of the recogniser than they would have been in a simpler task.

- Range of accuracy levels

  The range of recogniser performance levels tested in this experiment was considerably narrower than in the earlier one — even at the low performance level, the substitution rate on isolated digits was only 3%, with no rejections, whereas in the data entry experiment there was a digit error rate of at least 7.7% in the entered data, even after rejection and repetition of some inputs, in each of the imperfect recognition conditions. Of course the error rates on some of the other vocabularies in the banking experiment were worse than on the single digits, but participants may not have expected such high accuracy in these more demanding recognition tasks.

Further analysis of the first-use attitude results in relation to the actual number of recognition errors experienced by each participant confirmed that moderate numbers of errors (up to three in the course of the call) had no detectable effect on attitude. The attitude scores were lower for those who had experienced larger numbers of errors (four or more), but since these occurred for only five of the 120 participants it is impossible to draw firm conclusions without a further experiment.

One particular aspect of the participants' dialogue experiences that may have obscured the effect of accuracy is that the vocabulary coverage of the recogniser was inadequate — in particular, it excluded the use of 'double' in digit strings and did not accept 'correct' as a synonym for 'yes'. The resultant recognition failures — occurring indiscriminately at all levels of within-vocabulary accuracy — would tend to dilute the effect of the accuracy differences on utterances within the vocabularies. The effect of within-vocabulary accuracy differences should still be detectable given a large enough sample; but it would be better to improve the coverage of the vocabularies, both for the sake of a usable service and as a means of making experiments on accuracy and attitude more sensitive.

5. Conclusions

A novel, acoustically guided Wizard of Oz simulation technique has been developed and has been applied to the usability evaluation of spoken-dialogue systems incorporating a word-spotting recogniser. The simulation is more realistic than previous ones in that the recognition accuracy depends on the acoustic characteristics of the input, and, in particular, isolated and complete keywords are recognised more accurately than embedded and incomplete ones.

The experiments conducted have yielded a useful database of utterances obtained in response to varying styles of prompt messages in the context of a simulated telephone banking service. Notable differences were observed in the content and style of input utterances when the prompting style was varied. Such results can inform the design of dialogue messages and the specification of recognition vocabularies and grammars to be used with them.

It was found that users responded very differently to the different styles of prompt wording. When prompted specifically for a service name, they tended to respond with a single word or a short phrase, whereas after a more open-ended prompt such as ‘How can I help you?’ most users gave a sentence-length response. When the option of saying ‘help’ was mentioned in the prompt, many users did so immediately. Despite these very different spoken responses, there was little difference in the total number of inputs required to accomplish a series of tasks, or in users’ attitudes to the service.

The second experiment surprisingly revealed no difference in attitude between participants experiencing higher (100%) and lower levels of speech recognition accuracy. However, it is possible that this was due simply to the small size of the accuracy effect, relative to the level of random variations (due to differences in the details of individual participants’ dialogue experiences) occurring in a service of this complexity. The analysis by number of errors experienced suggests that the attitude measure may be sensitive only to large numbers of recognition errors (e.g. more than three in one call), but this cannot be established reliably from the present data since few participants experienced large numbers of errors. Further trials would be necessary to elucidate this issue, and to determine whether the insensitivity to a small number of errors observed with the attitude questionnaire in an experimental setting extends to satisfaction with services in real-world use.

One source of problems in the dialogues, unrelated to the accuracy level of the recogniser, was the deficiency in the coverage of the recognition vocabularies, whereby some utterances were not recognised successfully because they were outside the predefined vocabulary at the point in the dialogue where they occurred. In further experiments, and in the design of real services, it will be very important to ensure that the vocabulary or grammar at each point in the dialogue covers all likely ways in which the user may give the requested information.

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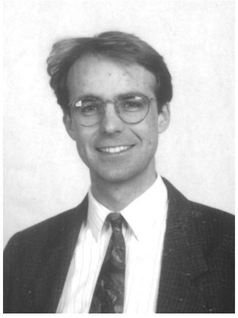

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